

MATH 30803

Linear Algebra

Fall 2024

Professor John Bergdall
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MWF 12:55 - 1:45 PM
Classroom: WJWH 302

Overview

This course studies linear algebra from the ground up. We start by explicitly solving linear equations. Then, we reimagine the concepts using vector spaces. Those abstractions lead to clarity in major applications, including data fitting (used in Internet searches) and data compression (how your devices store so many photos).

Learning goals

Solve (large!)
systems of linear
equations

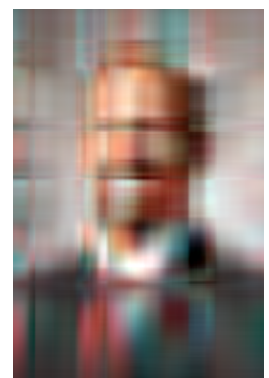
Learn matrix
algebra and
vector spaces

Apply to data
fitting and
compression

Explore
orthogonality &
eigenconcepts

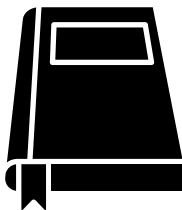
Try it out!

I compressed my portrait using
singular value decomposition. See
<https://timbaumann.info/svd-image-compression-demo>



Background top image: <https://mathworld.wolfram.com/LeastSquaresFitting.html>

Basic resources



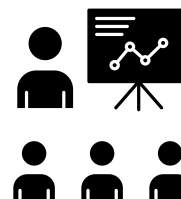
Textbook

Linear algebra with applications
(10th ed.) by Leon and de Pillis
(Rent digital/print: \$44/\$86*)



Course materials

Materials posted to Blackboard.
Assignments on Gradescope.
(Free)



Office hours (SCEN 220)

Mon: 2:00 - 3:00
Thu: 2:30 - 3:30
Fri: 9:30 - 10:30

*If costs cause hardship or would limit your course access, email me for support.

Tasks and grades

Course tasks are divided into *Supporting Tasks* and *Performance Tasks*. Supporting Tasks represent class participation and readings. Performance Tasks represent chances to show your understanding of concepts and techniques. Further details on pages 3-4.



Supporting Tasks

Class Participation and Reading
Quizzes



Performance Tasks

Exams, Homework Sets and an
Application Portfolio

Turn-in

Reading Quizzes are due at the *start of class* (12:55 PM). **Homework Sets** are typically due *Monday night at 11:59 PM*.

Revisions

Revising your work reinforces your learning. Except for a final exam, every Task may be revised *once*.

- Revisions for Reading Quizzes are due **Mondays at 11:59 PM**. (Same time as Homework Sets.)
- Revisions for Homework Sets will be collected roughly **36 hours** after they are graded and returned.

Late Tasks

Completing Tasks by their due dates keeps you on track. Still, you *may turn work in late as part of the revision process*. However, you lose your chance to revise the work. **It is to your benefit to turn in partially completed work** even if it is not perfect.

Supporting Tasks

Class participation gives you voice and agency in your learning. **Reading Quizzes** measure understanding of readings that frame class discussions.



Class Participation (1 point per Task)

- **Class surveys.** Three (3) surveys throughout the semester.
- **Class welcomes.** Two (2) meeting summaries read aloud to start class.

Details. One way students support learning is structural participation in the course.

- **Class surveys.** At the start, middle, and end of the semester, I survey the course to reinforce what is working and address what is not. The first survey is not anonymous, but the remainder are. You will earn your point by affirming you participated in the survey.
- **Class welcomes.** A class welcome is a short 1-2 paragraph summary of the previous class. You will work with a partner to write the summary, post it to Blackboard, and then read it aloud in the first 2-3 minutes of the next meeting. Sign-ups will be made in the first weeks of class.

Reading Quizzes (1 point per question)

- **2-3 questions** per reading.
- **Initial responses** in timed and auto-graded quiz on Gradescope.
- **Revisions** will include complete reasoning, in writing.



Details. Reading Quizzes questions are *multiple choice* or *select all*.

- Before the quiz, you will know a rough form of the questions but not the answer options.
- You take the quiz prior to a class meeting and the questions frame the class discussion.

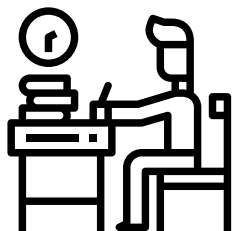
If you do not get a question initially correct, then revision is possible! Your revision will be writing a detailed response that identifies the correct answer *and* explains the reasoning.

Revision specifications. Here is how you will “revise” your Reading Quizzes. **See page 12 for an example.**

- **Multiple choice questions.** Identify the correct answer along with an explanation for why it is correct.
- **Select all questions.** First, identify the correct and incorrect options. Second, explain why the correct ones are correct *and* why the incorrect ones are incorrect.
- Complete and accurate revisions **earn back full points**. Partial explanations may be awarded zero or half a point, based on my discretion.

Performance Tasks

Homework Sets are first shots at concepts and techniques. **Mastery Assessments** measure broad proficiency. Your **Application Portfolio** brings your learning into contact with the broader scientific world.



Homework Sets

- **A few questions** per section of Leon-de Pillis, due Monday nights.
- **Turned in** electronically through Gradescope.
- **Collaborate!** As long as you attribute credit to your source(s).

Mastery Assessments (exams)

- **In-class exams** focused on vital learning goals and skills.
- **In-class midterms** on 09/20 and 10/18. **Final exam** 12/09, 12:45-2:45.
- **5-6 questions** per hour, with a mix of calculation and concepts.



Application Portfolio

- **Major work** during Units 3 and Unit 4.
- **Detailed prompts** focused on applications in the text.
- **Opportunity** to pursue material with choice based on your interests.

Revision specifications. Performance Task revisions are due at the first 11:59 PM on a Mon-Thu no earlier than **36 hours after the initial grades** are returned. (I will announce these deadlines in class and on Blackboard. Do not stress about the complex rule.) The method of revision depends on the Task.

Homework Sets and the Application Portfolio. To revise a question on a Homework Set or your Application Portfolio, you have two options.

- *Option #1.* Completely rewrite the new solution to the problem. We will assess the work as if we never saw the prior answer. Your new grade will be your final grade.
- *Option #2.* Explain in writing the mistakes you made and how to correct them. We will reassess the work by taking your explanation into account. Your adjusted grade will be the final grade.

Mastery Assessments. Revising work on a Mastery Assessment is a two-step process.

- *Step #1.* Explain in writing the mistakes you made and how to correct them. (Due in the 36-hour window.)
- *Step #2.* If your explanations are accurate, I will give you a second exam (schedule TBD) comprised of problems similar to the ones you want to revise. Your new marks will replace your old ones.

Performance Task Grading

Your Performance Tasks will be graded using a 6-point rubric. The rubric will be strictly applied because revisions are always possible. **Examples of how this rubric is applied are on pages 10-11.**

Presentation (2 points possible). Excelling in my 3xxxx-level courses requires understanding context and applying knowledge while presenting your answers in a clear and easy to understand fashion. Therefore, we provide two points possible for the presentation of your answer. Note: these points are theoretically independent of one another.

Presentation (up to 2 points)	Presentation #1	Presentation #2
	The work is neatly written, so the grader can read without difficulty.	Opening or closing sentences indicate the goals and outcome.

Remaining points (4 points possible). The bulk of the points are awarded for conceptual or strategic understanding (e.g. choosing an accurate theory to apply) and for mechanical and technical accuracy (e.g. performing calculations) and. We will rate you in both aspects on a scale of 0-2 points.

	0 points	1 point	2 points
Concepts or strategy (up to 2 points)	The answer misidentifies or misuses crucial concept(s) or applies unproductive strategies.	The answer identifies and uses a crucial concept or attempts a productive strategy.	The answer identifies or applies <i>all</i> crucial concepts and succeeds in carrying out a productive strategy.
Mechanics or technique (up to 2 points)	The answer has significant or recurring issues with its mechanics or technique.	The answer shows proficiency in some mechanics and techniques.	The mechanics and techniques in the answer are flawless (and of the intended difficulty level*).

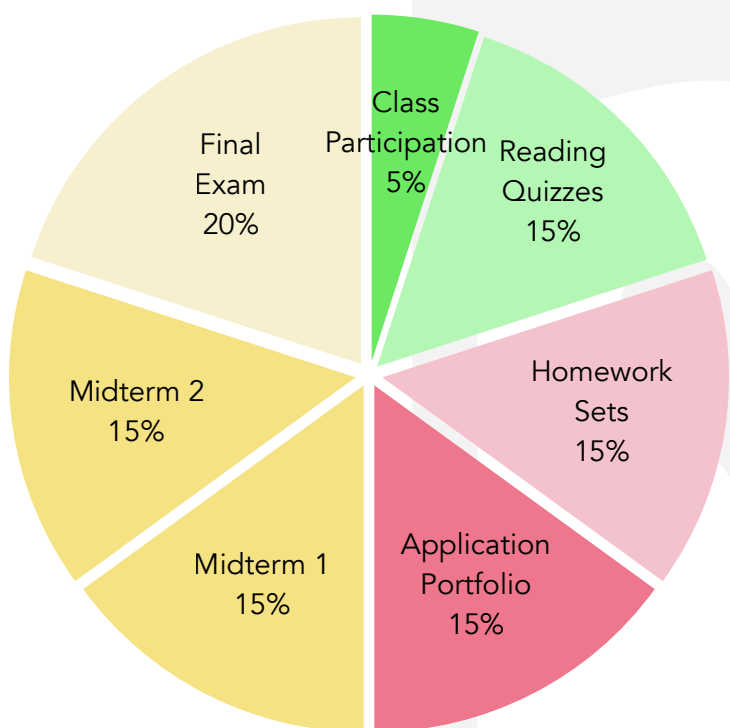
* "...the intended difficulty level" means you will **not** earn 2/2 if you perform a completely accurate calculation that is far simpler than the intended one. **See Solution #4 in the example on page 11.**

Caveat regarding the final exam. There is no revision possible on the final exam. So, "...all crucial..." and "flawless" in the 2-point criteria above will be suitably relaxed during the final exam.

Final grade

Your Task scores will be weighted according to the percentages in the pie chart below and to the left. Those weighted scores will then determine your final grade according to the table on the right. *I reserve the right to make the gradelines more lenient.*

Task-by-Task weights for final grade



Gradelines

Grade	Minimum weighted score
A	90%
B	80%
C	70%
D	60%

Rubric interpretation. The Performance Tasks are graded on a 0-6 scale and yet the gradelines are on a 90/80/70 scale. So, let me interpret your Task grades for you.*

- **A-level Performance Tasks = average 5.4.** So, half the time your work is excellent (earning 6's) and the other half there is at least excellence in concepts or mechanics (earning 5's).
- **B-level Performance Tasks = average 4.8.** So, nearly all the time your work is at least excellent in mechanics or concepts (earning 5's).
- **C-level Performance Tasks = average 4.2.** So, your work shows proficiency in mechanics and concepts (earning 4's).

* This assumes you present your work well (earning 2 points there) and it ignores any impact of the Class Participation and Reading Quizzes on your grade.

Tips for success

This class is designed for you to pass. The revision process puts you in control of your grade. Everyone will need some help at some point in the semester. So, here are a few tips and resources you can pursue.

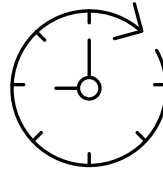


Message!

Write for help.



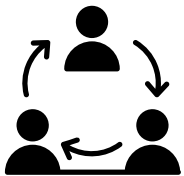
Ask! Me early and often.



Budget! Take your time.



Read! The text is a guide.



Collaborate!

With peers.

More tips.

- Take a break. Walk away.
- Explain to someone why you are stuck.
- Work out a simpler example.
- Search the internet for key terms.
- Keep going!
- Tutoring is available in the Student Success Center in CORD.

Further Resources

Appointments

Office hours will not always fit your schedule. Email me to set appointments. In your initial message, please explain (i) what you want to talk about, (ii) *your* availability and (iii) if meeting in person is important (Zoom is possible).



Email

Your questions are important. Email me or post to Blackboard. (I may ask you to do this anyways.) Responses may take 24 hours, but send a reminder if I do not reply.

Access

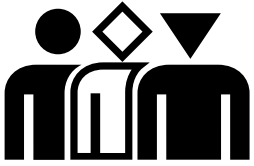
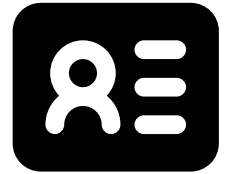
I want you to pass this class, with success. If you find resources not being provided, tell me and I will work with you to fix the issue. For some students, the Center for Educational Access (cea.uark.edu) coordinates student accommodation requests. Please request your accommodation letter early semester and meet with me to discuss it.



Class formation

Names and pronouns

You deserve to be called how you want. Let me know your preferred name and pronouns any time. When meeting one another, re-introduce yourself *frequently*. Everyone's names, pronouns, etc. should be properly used.



Diversity

I pledge attention to your identity and experience, regardless of age, background, beliefs, ethnicity, gender identity and expression, national origin, racial identity, religious beliefs, sexual orientation, and any other visible or non-visible categories. You are expected to do the same for me and your peers.

Academic integrity

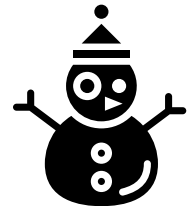
I trust my students to follow (i) the University of Arkansas Academic Integrity Policy and (ii) my own explicit assignment instructions. When you have questions about a boundary or rule, ask me rather than assuming you know the answer.



Technical policies

Inclement weather

If the University closes due to inclement weather, our course will meet online. The meeting will be recorded, as dictated by University policy. Any assessment (i.e. exam) or due date will be postponed until the next day the University is open.



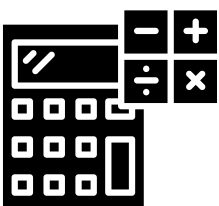
Class cancellation (emergencies)

If class is cancelled on short notice, I will work with the math office to try to find a substitute. If a substitute is found, class will be held. If not, alternative learning materials and assessments will be supplied. Regardless, an announcement will be sent promptly via email and posted to Blackboard.



Devices

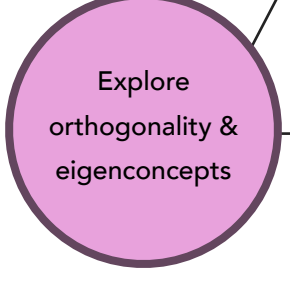
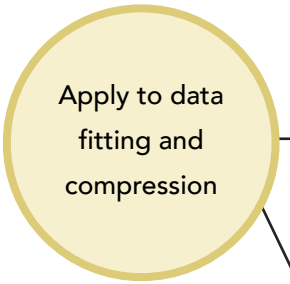
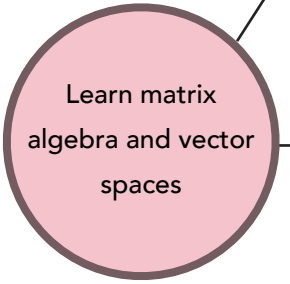
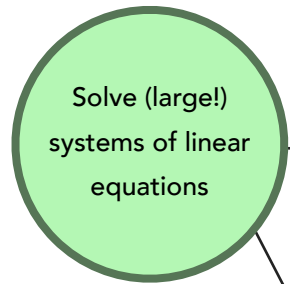
I will record class meetings using in-room recording systems. If you use your devices (phone, tablet, computer, etc.) to participate in meetings, you are expected to use those devices productively. Non-course activity distracts your peers from learning and me from teaching. It will not be tolerated.



Calculators

A simple non-programmable calculator is not required. However, they are allowed and may be useful to give you confidence in your calculations during exams.

Schedule



Unit	Wk.	Dates	Topic(s)	Reading (L.-dP)
1	1	Aug 19-23	Systems of linear equations, solutions by row reduction algorithms.	Sec. 1.1-1.2
	2	Aug 26-30	More row reduction, matrix arithmetic and algebra	Sec 1.2-1.4
	3	Sep 02-06	Matrix algebra and elementary matrices	Sec 1.4-1.5
	4	Sep 09-13	Determinants, vector subspaces	Sec 2.1-2.2, 3.2
2	5	Sep 16-20	Subspaces, review and midterm #1	Sec 3.2
	6	Sep 23-27	Spanning sets, linear independence, and bases	Sec 3.3-3.5
	7	Sep 30-Oct 04	Fundamental subspaces, intro to linear transformations	Sec 3.6, 4.1
	8	Oct 07-11	Matrices and linear transformations, introduction to orthogonality	Sec 4.2, 5.1-5.2
3	9	Oct 14-18	Fall break, review and midterm #2	—
	10	Oct 21-25	Introduction to least squares problems	Sec 5.2-5.3
	11	Oct 28-Nov 01	Advanced least squares theory	Sec 5.5-5.6
	12	Nov 04-08	Applications and intro to eigenvectors	Sec 5.6-6.1
4	13	Nov 11-15	Applications to differential equations	Sec 6.1-6.2
	14	Nov 18-22	Diagonalization and singular value decompositions	Sec 6.3-6.5
	15	Nov 25-29	Final applications, Thanksgiving break	Sec 6.5
	16	Dec 02-04	Review for final exam	—

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Performance Task Example

Here we provide an example for how the rubric for grading a performance task might be applied. We show a problem from calculus, hoping that the general idea is familiar enough to you.

Prompt/Question.

Determine the area between the curve $y = \sin(x)$ and $y = \cos(x)$ over the x -range of $[0, \pi/4]$.

Solution #1.

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = -\sin(x) - \cos(x) \Big|_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414\ldots$$

Total score: 5/6.

- *Presentation* — 1/2. The writing is neat (it is typed after all) but there are no sentences indicating the goal or outcome of the problem.*
- *Concept/strategy* — 2/2. The solution identifies the correct integral to calculate.
- *Mechanics/technique* — 2/2. The integral is calculated completely accurately.

* This illustrates an important point in the rubric. The problem asked for an area. Part of your work should reflect your understanding that finding areas is converted (via calculus) into calculating integrals!

Solution #2. We find the area by calculating an integral. Since $y = \cos(x)$ lies above $y = \sin(x)$ on $[0, \pi/4]$, the relevant integral is:

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx.$$

Now we calculate:

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = -\sin(x) + \cos(x) \Big|_0^{\pi/4} = 1.$$

So, the area between the curves is 1.

Total score: 5/6.

- *Presentation* — 2/2. The writing is neat and the opening and closing sentences explain the goal and outcome.
- *Concept/strategy* — 2/2. The solution identifies the correct integral to calculate.
- *Mechanics/technique* — 1/2. The integral is not calculated accurately. It (seemingly) misidentifies the anti-derivative of $\sin(x)$.

PT Example, cont.

Here we provide an example for how the rubric for grading a performance task might be applied. We show a chose a problem from calculus, hoping that the general idea is familiar enough to you.

Prompt/Question.

Determine the area between the curve $y = \sin(x)$ and $y = \cos(x)$ over the x -range of $[0, \pi/4]$.

Solution #3. We find the area by calculating the integral

$$\int_0^{\pi/4} \sin(x) - \cos(x) \, dx = 1 - \sqrt{2} \approx -0.414\dots$$

Since areas are positive, the answer is 0.414...

Total score: 5/6.

- *Presentation* — 2/2. The writing is neat and the opening and closing sentences explain the goal and outcome.
- *Concept/strategy* — 1/2*. The solution identifies that an integral should be calculated, but it misidentifies which integral.
- *Mechanics/technique* — 2/2. The given integral is accurately calculated and then the answer is made positive, which is a good sign (lol!).

* The way the solution is written, I assume the misidentification of the integral is a strategic error. If the student wrote "... $y = \sin(x)$ is above $y = \cos(x)$ on $[0, \pi/4]$..." then I would give 2/2 for strategy. But, I would have also switched to 1/2 for mechanics since of course $y = \sin(x)$ actually lies *below* $y = \cos(x)$ on $[0, \pi/4]$.

Solution #4. The area is given by

$$\sin(\pi/4) - \cos(\pi/4) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0.$$

Total score: 2/6.

- *Presentation* — 1/2. The writing is neat, but I have an issue with the lack of closing sentence. In this case the student found the area was zero. That deserves to be addressed.
- *Concept/strategy* — 0/2. The solution strategy is a calculation seemingly unrelated to the area question.
- *Mechanics/technique* — 1/2*. The calculation is literally correct. However, it is far simpler than the intended calculation (no anti-derivatives).

* This illustrates that the rubric categories are not completely independent. Failing to identify the strategy can lead to a gap in mechanical proficiency.

Reading Quiz Example

Revising *select all that apply* questions requires explaining both the logic behind the “right” answers and the logic behind the “wrong” ones. Let us see how that looks in practice.

Question. (Select all that apply) Which of the following indefinite integrals are calculated accurately?

(a) $\int \cos(x) dx = \sin(x) + C.$

(b) $\int \sin(x) dx = \cos(x) + C.$

(c) $\int \tan(x) dx = \sec^2(x) + C.$

Sample “revision”. The integral (a) is accurate but (b) and (c) are not. The answer (a) is true because $\cos(x)$ is the derivative of $\sin(x)$ and so $\sin(x)$ is the anti-derivative of $\cos(x)$. Next, (b) is false because it should say $\int \sin(x) dx = -\cos(x) + C$. And, (c) is false because, in fact,

$$\int \tan(x) dx = \ln(|\sec(x)|) + C.$$

Tip. To show something is *false*, you can provide a corrected statement.

Question. (Select all that apply) Suppose $y = f(x)$ and $y = g(x)$ are two curves lying over x in $[a, b]$. Which of the following statements are accurate?

(a) The area between the two curves is always non-negative.

(b) The integral $\int_a^b f(x) - g(x) dx$ represents the area between the two curves.

(c) The integral $\int_a^b f(x) - g(x) dx$ is always non-negative.

(d) The integral $\int_a^b |f(x) - g(x)| dx$ represents the area between the two curves.

Sample “revision”. The answers (a) and (d) are accurate, while (b) and (c) are not. For (a), areas are always non-negative so that is easy. For (d), this integral is how you calculate the area (we learned this in the reading on page 109). Now we deal with (c) and (b). Suppose $f(x) = x$ and $g(x) = 2$ over the interval $[0, 1]$. Then

$$\int_0^1 x - 2 dx = \frac{1}{2} - 2 = -1.5.$$

So, (c) fails. The main issue is $g(x) = 2$ lies above $f(x) = x$ on $[0, 1]$ and so $\int_0^1 f(x) - g(x) dx$ calculates the *negative* of the area in this case. This explains why (b) is false.

Tip: To show something is *false*, you can also provide a counter-example like we do here for cases (b) and (c).

Tip: To show something is *true*, you can reference your reading or class discussion, as long as you are specific!